

Global Supply Chain and Risk Analysis

Data Analytics Project using Power BI

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Project Introduction & Objective

- This project focuses on analyzing global shipment operations to understand performance trends, operational efficiency, and risk factors affecting supply chain activities.
- The objective of this project is to monitor shipment performance, identify disruption risks, and support data-driven decision-making through interactive dashboards and data visualization.

Dataset Source & Overview

Data Source: Kaggle

The dataset used in this project is a simulated global supply chain dataset designed for analytical and learning purposes.

- **Total Records:** 5,000 shipments
- **Data Type:** Operational and logistics data
- **Scope:** Global transportation and shipment activities
- **Data Structure:** Transactional shipment records

The dataset contains information related to shipment locations, product categories, transportation conditions, and disruption indicators.

Important Columns & Data Types

- **Shipment_ID** -Text
- **Product_Category** - Text
- **Origin_Port** -Text
- **Destination_Port** - Text
- **Transport_Mode** -Text
- **Lead_Time_Days** - Decimal Number
- **Weight_MT** - Decimal Number
- **Distance_km** - Decimal Number
- **Weather_Condition** - Text
- **Fuel_Price_Index** - Decimal Number
- **Geopolitical_Risk_Score** - Decimal Number
- **Disruption_Occurred** - Whole Number (0 = No, 1 = Yes)

These columns were selected because they directly support operational and risk analysis.

Data Cleaning & Transformation

- Verified the dataset for duplicate records
- Checked for missing or null values
- Ensured correct data types for each column
- Validated data consistency across fields
- Prepared the dataset for analysis and reporting

Data Modeling

- Created a Fact table to store shipment transaction data
- Developed Dimension tables for Product, Port, Transport, Weather and Date information
- Established relationships between tables using key fields
- Implemented a star schema data model

This data model improves performance, supports accurate filtering, and enables efficient analysis.

DAX Measures & Calculations

Operational Metrics

- Total Shipments
- Total Distance
- Total Weight
- Average Lead Time
- Total Product Categories
- Total Ports

Performance Metrics

- Average Reliability Score
- Most Reliable Product Category
- Most Reliable Port

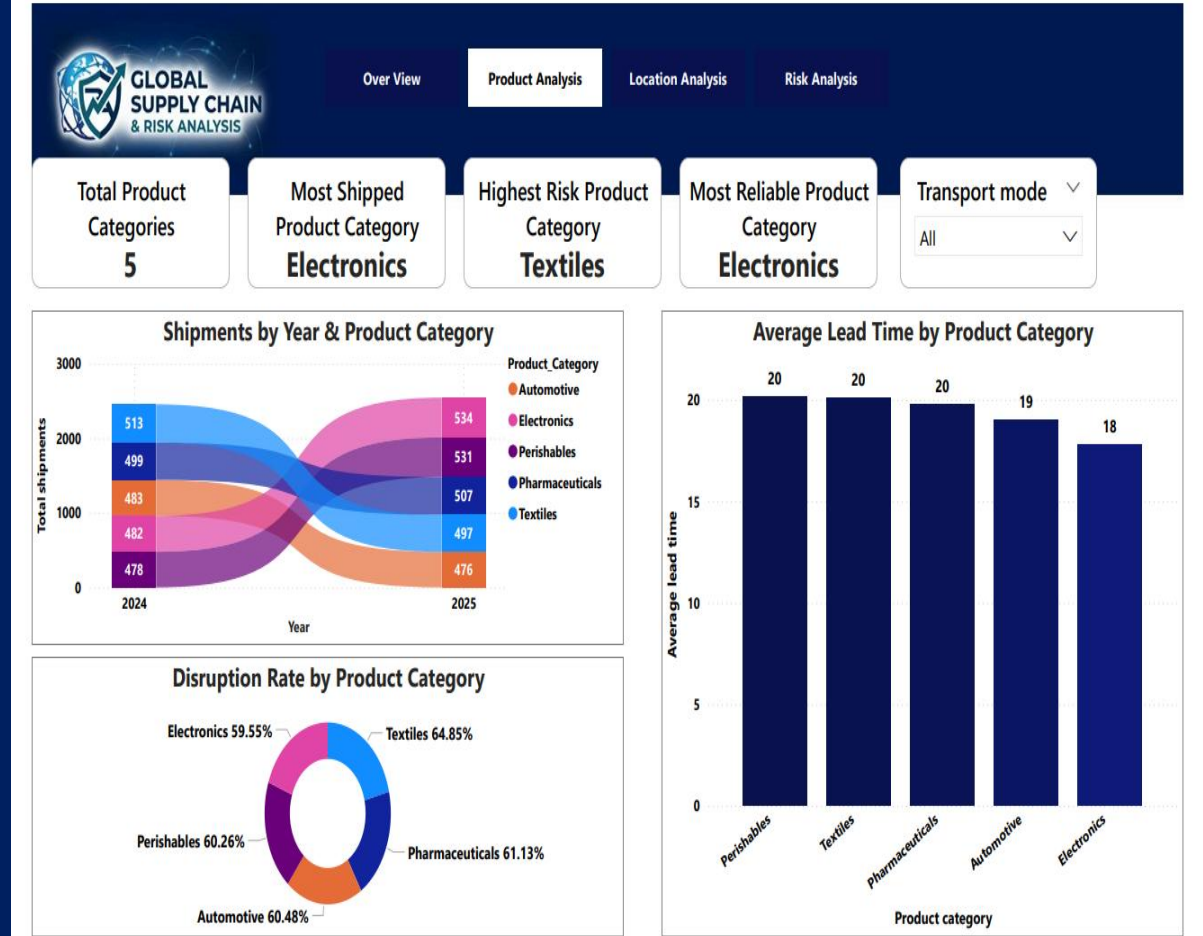
Risk Metrics

- Disruption Rate
- Average Disruption Rate
- Total Disrupted Shipments
- Average Geopolitical Risk Score
- Highest Risk Product Category
- Weather Condition with Highest Disruption
- Product Category with Highest Disruption

Logistics & Flow Metrics

- Most Shipped Product Category
- Top Exporting Port
- Top Importing Port

Report Design & Visual Appearance





- Designed a clean and professional dashboard layout
- Used consistent color themes and formatting across all pages
- Selected appropriate chart types for each type of analysis
- Included KPI cards for quick performance monitoring
- Applied clear titles and labels to improve readability

Power BI Features Used

- Slicers implemented across report pages, including Shipment Start (Origin Port), Shipment End (Destination Port), Transport Mode, Date Hierarchy (Year, Quarter, Month, Day), and Year.
- Interactive filters for dynamic data exploration.
- Drill-down functionality for detailed analysis.
- Tooltips to display additional information.
- Cross-filtering between visuals.
- Page Navigator for easy navigation between dashboard pages.

Key Analysis and Insights

Shipment Patterns

- The highest number of shipments occurred during fog weather conditions (20.7%), followed by storms.
- Air transport mode handled the highest shipment volume (1,320 shipments), followed by sea transport.
- Electronics was the most shipped product category, followed by textiles.

Product Performance

- Textiles recorded the highest disruption rate (64.85%), followed by automotive products.
- Perishables, textiles, and pharmaceuticals showed the highest average lead times.

Location Performance

- Busan was the busiest origin port (667 shipments), followed closely by Rotterdam.
- Shanghai and Rotterdam recorded the highest average lead time among origin ports (21 days).
- Busan, Singapore, and Marseille showed the highest average lead time among destination ports (21 days).

Risk and Disruption Analysis

- Hurricane conditions resulted in the highest disruption rate.
- Air transport mode showed the highest disruption rate.
- A total of 3,063 shipments were disrupted.
- Geopolitical risk scores were analyzed on a quarter-wise basis, with peak values observed in February, April, July, November and December.

Storytelling, Business Impact, and Conclusion

Storytelling

This project analyzed global shipment operations to identify patterns, operational efficiency, and risk factors affecting supply chain performance.

The dashboard enables monitoring of shipment trends, disruptions, and delivery performance across products, locations, and transportation modes.

Business Impact

- Identify high-risk weather conditions and reduce disruptions
- Improve transportation planning and delivery schedules
- Monitor shipment performance using real-time metrics
- Support data-driven decision-making

Conclusion

The Global Supply Chain & Risk Analysis dashboard demonstrates how data visualization and analytics can improve operational visibility, risk management, and supply chain efficiency.

Thank You All